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Gao Yuman

EDUCATION

Zhejiang University

Sep 2021-Jun 2026

- Ph.D. of Control Science and Engineering (CSE), FAST Lab
- Advisor: Prof. Fei Gao

University of California, Berkeley

Nov 2024-Dec 2025

- One-year Visiting Scholar (Ph.D.), Hybrid Robotics Lab
- Advisor: Prof. Koushil Sreenath, Prof. Yi Wu, Prof. Zhongyu Li

Zhejiang University

Sep 2017-Jun 2021

- CHU KO CHEN Honors College (**Top 8%** of students in the whole school):
Advance Honor Class of Engineering Education (**Top 2%** of students in all engineering disciplines)
- BSc of Control Science and Engineering, Robotics
- **GPA: 3.96/4.00, Rank: 1/57**
- Core modules: Optimization (4.0/96), C++ (4.0/95), Data Structures (4.0/95), Physics (4.0/96), Numerical Computation (4.0/95), Big Data Analysis (4.0/95)

My primary research interests lie in Motion Planning, Navigation, Autonomous Exploration, Reinforcement Learning, Multi-robot System. Published 2 IEEE T-RO papers (1 first-author), 5 Science Robotics papers. 5 First/Co-first-author papers in total;

PUBLICATIONS

(*Equal Contribution, †Corresponding Author)

- [1] **Y. Gao**, J. Ji, Q. Wang, et al, S. Shen, C. Xu, and F. Gao. "Adaptive Tracking and Perching for Quadrotor in Dynamic Scenarios." *IEEE Transactions on Robotics* (**T-RO**), 2024.
- [2] **Y. Gao**, R. Zhang, T. Lai, Y. Cao, C. Xu, and F. Gao. "Autonomous Exploration With Terrestrial-Aerial Bimodal Vehicles." *IEEE Robotics and Automation Letters* (**RA-L**), 2025.
- [3] **Y. Gao**, Y. Wang, et al, C. Xu, and F. Gao. "Meeting-merging-mission: A multi-robot coordinate framework for large-scale communication-limited exploration." *IEEE/RSJ International Conference on Intelligent Robots and Systems* (**IROS**), 2022.
- [4] Z. Su*, **Y. Gao***†, E. Lukas, Y. Li, J. Cai, F. Tulbah, F. Gao et al. "Toward Real-World Cooperative and Competitive Soccer with Quadrupedal Robot Teams." *Conference on Robot Learning* (**CoRL**), 2025.
- [5] R. Jin*, **Y. Gao***, Y. Wang, Y. Wu, H. Lu, C. Xu, and F. Gao. "Gs-planner: A gaussian-splatting-based planning framework for active high-fidelity reconstruction." *IEEE/RSJ International Conference on*

- [6] H. Liu, **Y. Gao**, S. Teng, Y. Chi, Y. Shao, Z. Li, M. Ghaffari, K. Sreenath. "Ego-Vision World Model for Humanoid Contact Planning." *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [7] X. Zhou, X. Wen, Z. Wang, **Y. Gao**, et al. "Swarm of micro flying robots in the wild." *Science Robotics* 7, 2022.
- [8] Q. Liao, TE. Truong, X. Huang, **Y. Gao**, G. Tevet, K. Sreenath, CK. Liu. "Beyondmimic: From motion tracking to versatile humanoid control via guided diffusion" *Science Robotics*, 2026.
- [9] Q. Wang, **Y. Gao**, J. Ji, C. Xu, and F. Gao. "Visibility-aware trajectory optimization with application to aerial tracking." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021.
- [10] M. Wang, Q. Wang, Z. Wang, **Y. Gao**, J. Wang, C. Cui, Y. Li et al. "Unlocking aerobatic potential of quadcopters: Autonomous freestyle flight generation and execution." *Science Robotics* 10, 2025.
- [11] B. He, Z. Wang, Y. Zhou, J. Chen, C. Singh, H. Li, **Y. Gao** et al. "Microsaccade-inspired event camera for robotics." *Science Robotics* 9, 2024.
- [12] Z. Han, M. Tian, Z. G., D. Xue, J. Xing, Q. Wang, **Y. Gao**, J. Wang, C. Xu, and F. Gao. "Hierarchically depicting vehicle trajectory with stability in complex environments." *Science Robotics* 10, 2025.
- [13] Q. Wang, Z. Wang, M. Wang, J. Ji, Z. Han, T. Wu, R. Jin, **Y. Gao**, C. Xu, and F. Gao. "Fast iterative region inflation for computing large 2-D/3-D convex regions of obstacle-free space." *IEEE Transactions on Robotics (T-RO)*, 2025.
- [14] Y. Shao, X. Huang, B. Zhang, Q. Liao, **Y. Gao**, Y. Chi, Z. Li, S. Shao, and K. Sreenath. "LangWBC: Language-directed Humanoid Whole-Body Control via End-to-end Learning." *Robotics: Science and Systems (RSS)*, 2025.
- [15] T. Sheng, R. Jin, C. Yang, K. Qiu, M. Wang, J. Shi, J. Zhang, **Y. Gao** et al. "Unmanned aerial vehicle mediated drug delivery for first aid." *Advanced Materials* 35, 2023.
- [16] L. Wang, H. Ye, Q. Wang, **Y. Gao**, C. Xu, and F. Gao. "Learning-based 3d occupancy prediction for autonomous navigation in occluded environments." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021.
- [17] R. Zhang, J. Lin, Y. Wu, **Y. Gao**, C. Wang, C. Xu, Y. Cao, and F. Gao. "Model-based planning and control for terrestrial-aerial bimodal vehicles with passive wheels." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
- [18] Z. Su, B. Zhang, N. Rahmanian, **Y. Gao**, Q. Liao, C. Regan, K. Sreenath. "Hitter: A humanoid table tennis robot via hierarchical planning and learning" *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

RESEARCH EXPERIENCES

Reinforcement Learning on Legged Robots

2024-2025

UC Berkeley, Hybrid Robotics Lab, Advised by Prof. Koushil Sreenath, Prof. Yi Wu and Prof. Zhongyu Li

[Learning Quadruped Robot Team Soccer]

Multi-agent reinforcement learning for the cooperative and competitive multi-agent soccer game.

Demonstrate, for the first time, a fully decentralized multi-quadruped robotic system capable of playing soccer in the real world.

(Publication: [4] *CoRL* 2025)

[Beyondmimic: Humanoid Motion Control]

A unified motion imitation framework that enable humanoid robots to learn versatile, robust, and generable whole-body motion from human data. Achieving versatile humanoid motion control via latent diffusion model.

(Publication: [8] *Science Robotics* 2026)

[Ego-vision World Model for Humanoid Contact Planning]

Learn a visual world model that scalably captures the dynamics of diverse contact tasks, trained entirely on a demonstration-free offline dataset. Introduce a sampling-based MPC framework using a learned surrogate value function to guide contact planning process.

(Publication: [6] *ICRA* 2026)

Autonomous Exploration and Active Reconstruction

2022-2025

Zhejiang University, Fast Lab, Advised by Prof. Fei Gao

[Autonomous Exploration with Bimodal Vehicles]

Develop a complete autonomous exploration system for terrestrial-aerial bimodal robots, enabling flexible decision-making (including modality switch) under time and energy constraints. Implement multi-layer terrain perception, bimodal Monte Carlo Tree Search (MCTS) planning, and trajectory generation under bimodal dynamics constraints.

(Publication: [2] *RA-L* 2025)

[Gaussian-Splatting-based Active High-fidelity Reconstruction]

Active photorealism scene reconstruction with quadrotor utilizing 3D Gaussian Splatting technique.

Online model-consistent completeness and quality evaluation. A planning framework for active reconstruction and safely navigation with 3DGS as scene representation.

(Publication: [5] *IROS* 2024)

Motion Planning and Trajectory Optimization

2022-2024

Zhejiang University, Fast Lab, Advised by Prof. Fei Gao

DJI Co., Planning & Control internship, Advised by Prof. Shaojie Shen

[Adaptive Tracking and Perching Trajectory Optimization in Dynamic Scenarios]

High-efficiency full-onboard safe and perception-aware trajectory optimization framework. Deploy on Commercial quadrotors with fully onboard computation; achieve high-speed dynamic tracking and agile

perching at 10 m/s and on 60° tilt surfaces.

(Publication: [1] *T-RO* 2024 (Highest Rank *No.5* Popular Paper), DJI *International Paten* First Author, *Commercialized* as the BYD “Lingyuan” Intelligent Vehicular Auxiliary Drone System)

Multi-Robot System

2021-2022

Zhejiang University, Fast Lab, Advised by Prof. Fei Gao

[Multi-robot Coordinate Framework for Large-scale Communication-limited Exploration]

Collaborative exploration of unknown region by robots with communication range limitation. Efficient free space representation and querying via a particular polyhedron star convex. Effective collaborative exploration strategy based on designed meeting rule.

(Publication: [3] *IROS* 2022)

[Swarm of Micro Flying Robots in the Wild]

Demonstrated a swarm of 10 fully autonomous, fully onboard-sensing, and fully onboard-computing palm-sized aerial robots safely navigating through dense bamboo forests; recognized by *Science Robotics* as the world’s first autonomous aerial swarm system operating in unstructured natural environments.

(Publication: [7] *Science Robotics* 2022)

SELECTED AWARDS

Zhejiang University Outstanding Student	2023
Zhejiang University Scholarship-First Prize (Top 3%)	2021
Zhejiang Province Outstanding Graduates (Top 5%)	2021
CHUNHUI Scholarship (Top 2%)	2021
Zhejiang Province Scholarship (Top 3%)	2020

COMPETITION EXPERIENCES

Mathematical Contest in Modeling, First Prize (Notional level, China) (Top 0.7%)	2019
Mathematical Contest In Modeling (U.S.), Meritorious Winner (Top 13%)	2019
World Robotic Sailing Championships (U.K.), Third Place	2018

SOCIAL WORKS

Zhejiang University Student Association of Unmanned Systems **President / Founder**

- Led the team to participate in four world-class robotics competitions (UK, Canada, China) and hosted one international robotics event
- Top-rated tech student organization on campus; awarded University Social Work Excellence

COLLABORATOR

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| • Zhongyu Li, UCB (RL on Legged Robot) | • Jiaze Cai, MIT (RL on Flapping Wing) |
| • Ruiqi Zhang, UCB (Aerial Robotics) | • Xin Zhou, HKUST (Swarm of Robot) |
| • Chao Yu, Tsinghua (Multi-agent RL) | • Yunfei Li, Tsinghua (RL) |